

Marine Bioinformatics Hackathon: Biodiversity, Genomics, and Nextflow Pipelines.

Venue: Nile University 29-31 October 2025

Day 1 (29- 10 -2025)	
Timeline	Lectures and practical sessions
9:00 – 9:30	Registration
10:00 - 10:15	Welcome Note (Dr. Asmaa Abushady) & AfricaBP Introductory Session- Room 367, UB1
10:15 – 11:15	Lec1. Biodiversity Loss in Marine Biology (Dr. Sameh) Room 367, UB1
11:15 – 12:15	Lec2. Next Generation technologies: With a Deep dive in Marine Biology 367, UB1
12:30 – 13:00	Coffee break - Appetisers
13:00 – 14:00	Lec3. Applications of Metagenomics in Marine Drug Discovery (Dr. Sameh) - Room 367, UB1
14:00- 15:00	Practical Session 1: Quick tutorial on Galaxy platform and illustration for the selected pipeline (Shaimaa and Assem) - Room 367, UB1
15:00- 16:30	Practical Session 2: Getting data, QC and Sequence alignment (Shaimaa and Assem) - Room 367, UB1

Day 2 (30 - 10 -2025)	
Timeline	Lectures and practical sessions
9:00 – 10:00	Lec1.Integrated metagenome-metabolome analysis reveals coral responses to ocean warming (Dr. Amin Roshdy) - Room 367, UB1
10:00 – 10:30	Practical Session3. Extraction of Taxonomic Information (Shaimaa and Assem) - Room 53, UB1
10:30 – 11:30	Practical Session4. Visualization (Shaimaa and Assem) - Room 127, UB1
11:30 - 12:30	Coffee break - Appetizers
12:30 – 14:00	Lec2. Biodiversity Genomics in the '-Omics' Era (Dr.Abdollah Sharaf) Dr.Abdollah Sharaf - Room 127 , UB1
14:00- 15:00	Hackathon : Groups division and Projects assigning 367, UB1
15:00- 16:30	Hackathon : Working on group projects 367, UB1

Day 3 (31- 10-2025)	
Timeline	Lectures and practical sessions
9:00 – 10:30	Workflow management with Nextflow
10:30 – 11:00	Coffee break - Appetizers
11:00 – 12:00	Introduction to nf-core
12:00 - 12:30	Prima Project activities (Dr.Asmaa Abushady) - Room 367, UB1
12:30 - 15:30	Project discussions and honoring best group
15:30 – 16:00	Closing Ceremony

Visit the links below to access workshop practical materials

<https://github.com/assemkadry/AfricaBP-Biodiversity-Workshop>

Please Note:

- Lectures' materials will be uploaded by the end of the workshop